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Theorem of the Cube & Theorem of the Square

Recall (See-Saw Principle)

X : irred complete var.

T : integral sch.

$L \in \text{Pic}(X \times T)$

Suppose $\begin{cases} L|_{X \times \{t\}} : \text{triv.} & \text{for all } t \in T. \\ L|_{\{x_0\} \times T} : \text{triv} & \text{for at least one } x_0 \in X \end{cases}$

$\Rightarrow L : \text{trivial}$

(pt) see my note last time

Thm of the Cube

$X \times Y \times Z$: product of three irred. complete var.
with chosen $x_0 \in X, y_0 \in Y, z_0 \in Z$.

"line bundle L on $X \times Y \times Z$ is trivial"
iff

" $L|_{\{x_0\} \times Y \times Z}, L|_{X \times \{y_0\} \times Z}, L|_{X \times Y \times \{z_0\}} : \text{trivial}$ "

lem (rigidity lemma)

X : complete var. Y, Z : alg. var.

Assume that a morph $f: X \times Y \rightarrow Z$ contracts

$X \times \{y_0\} \rightarrow \{z_0\}$ for some $y_0 \in Y$
to a pt $z_0 \in Z$.

Then f is a composition of the projection $p_2: X \times Y \rightarrow Y$
and a morphism $g: Y \rightarrow Z$.

$X \times Y \xrightarrow{f} Z$

$\downarrow p_2$
 Y

$\nearrow g$

$f(x, y) = g(y)$

$f(x_0, y)$

Moreover, $f(X \times \{y_0\}) = \{z_0\} = f(\{x_0\} \times Y) \Rightarrow f : \text{const.}$

Pf of rigidity lem

Define $g(y) := f(x_0, y)$ for some fixed $x_0 \in X$.

Let $z_0 \in U_Z \subset Z$ affine open neighborhood.

$F := Z - U_Z$: closed

Now X is complete, then p_Z is closed map.

Thus, we have

$$G := p_Z(f^{-1}(F)) \text{ : closed.}$$

Moreover $y_0 \notin G$, then $U_Y := Y - G$ is no empty open $\subseteq Y$.

Now $\forall y \in U_Y$

$$\begin{array}{ccc} X \times \{y\} & \longrightarrow & U_Z \\ \text{complete} & & \text{affine.} \end{array}$$

Fact

X : complete conn. Z : affine
 $\phi: X \longrightarrow Z$ regular
 $\Rightarrow \phi$: const.

Therefore $X \times U_Y \subset X \times Y$ open & dense

& $X \times U_Y \xrightarrow[g \cdot p_Z]{f} Z$ coincide (open in irred $\Rightarrow$ dense)
 Hart. Ex I.1.3

by Hartshorne Ex II 4.2, $f = g \cdot p_Z$ on $X \times Y$.

Ex II 4.2

X : reduced sch / S
 Y : separated sch / S
 $f, g: S$ -morph of X to Y which agrees on
 an open & dense subset of X , then $f = g$.

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Cor 1 of rigidity lem

X, Y : abelian var
 $f: X \rightarrow Y$ any morph
 $\Rightarrow f(x) = h(x) + a$ h : homomorphism of X .
 $a \in Y$

pf replacing f by $f - f(0)$ we may assume $f(0) = 0$.

claim f : homo.

Consider $X \times X \xrightarrow{\phi} Y$ $\phi(x, y) = f(x+y) - f(x) - f(y)$

Then $\phi(X \times \{0\}) = \{0\} = \phi(\{0\} \times X) = 0 \Rightarrow \phi \equiv 0$ on $X \times X$
 $\Rightarrow f$ is homo. $\#$

Cor 2 of rigidity lem

X : abelian group $\Rightarrow X$ is a commutative gp.

pf Apply above $-1: X \rightarrow X$ inverse
 $(xy)^{-1} = x^{-1}y^{-1} = y^{-1}x^{-1}$ $\#$

Proof of theorem of the cube

Step 1 (Reduction to the case when X is a smooth curve)

$\forall x \in X$, there exists an irred curve $C \subset X$ passing through x & x_0 .
 $\hookrightarrow$ [See Mumford p.54 lemma.]

and $\tilde{C} \rightarrow C$ is normalization of C

We assume Thm is true for line bundles on $\tilde{C} \times Y \times Z$.

$\Rightarrow$ pull back of L to $\tilde{C} \times Y \times Z$ is triv.

$\Rightarrow L|_{\{x\} \times Y \times Z}$: triv

$\Rightarrow L|_{\{x\} \times Y \times \{z\}}$: triv for all $(x, z) \in X \times Z$.

$L|_{X \times \{y_0\} \times Z}$: triv for $y_0 \in Y$ & Y is complete

by Sec-Saw principle, L is trivial $\triangleleft$ step 1.

Step 2 In the case when X is a curve.

Assume X is a complete curve.

Consider L as a family of line bundles $L_{y,z}$ on X parametrized by $Y \times Z$

$$\text{s.t. } \begin{cases} L_{y_0,z} : \text{triv} & \forall z \in Z \\ L_{y,z_0} : \text{triv} & \forall y \in Y \end{cases}$$

$\Rightarrow L_{y,z}$ have degree zero for $\forall (y,z) \in Y \times Z$

so our family gives a morphism.

$$f: Y \times Z \rightarrow \text{Pic}^0(X)$$

$$\text{since } f(\{y_0\} \times Z) = \{0\} = f(Y \times \{z_0\})$$

by rigidity lemma, $f(y,z) \equiv 0$

by see-saw principle, L is trivial

$\triangleleft$ step 2.

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Corollary

A : abelian var (w/ $m: A \times A \rightarrow A$)

$$\pi_i: A \times A \times A \rightarrow A \text{ proj}$$

$$m_{ij}: A \times A \times A \rightarrow A \text{ given as } m \circ (\pi_i, \pi_j)$$

$$m: A \times A \times A \rightarrow A$$

$$(a,b,c) \mapsto a+b+c \quad (\text{abuse...})$$

then

$$m^* L \otimes \pi_1^* L \otimes \pi_2^* L \otimes \pi_3^* L \simeq m_{12}^* L \otimes m_{23}^* L \otimes m_{13}^* L$$

Consider $M \simeq (\text{LHS}) \otimes (\text{RHS})^{-1}$

$$g: A \times A \rightarrow A \times A \times A \quad g(y,z) = (0,y,z)$$

$$\begin{aligned} \rightarrow g^* M &\simeq \underbrace{g^* m^* L}_{m_{23}^*} \otimes \underbrace{g^* \pi_1^* L}_{\text{triv}} \otimes \underbrace{g^* \pi_2^* L}_{\pi_2^*} \otimes \underbrace{g^* \pi_3^* L}_{\pi_3^*} \otimes \underbrace{g^* m_{12}^* L}_{\pi_2^*} \otimes \underbrace{g^* m_{23}^* L}_{m_{23}^*} \otimes \underbrace{g^* m_{13}^* L}_{\pi_3^*} \\ &\simeq \text{triv.} \Rightarrow M|_{\{0\} \times A \times A} = \text{triv.} \end{aligned}$$

Similarly, we have $M|_{A \times \{0\} \times A}, M|_{A \times A \times \{0\}} : \text{triv}$
 $\Rightarrow$ by thm of the cube, M is triv $\neq$

Let $f, g, h : X \rightarrow A$ morph from var X .
 Pull back M by $(f, g, h) : X \rightarrow A \times A \times A$, we have

$$(f+g+h)^*L \simeq (f+g)^*L \otimes (f+h)^*L \otimes (g+h)^*L \otimes f^*L^{-1} \otimes g^*L^{-1} \otimes h^*L^{-1}$$

Now we consider

$f \equiv x, g \equiv y, h = \text{id}_A$ then

$$\begin{aligned} t_{x+y}^*L &\simeq t_x^*L \otimes t_y^*L + (x+y)^*L \otimes x^*L^{-1} \otimes y^*L^{-1} \otimes \text{id}^*L^{-1} \\ &\simeq t_x^*L \otimes t_y^*L \otimes L^{-1} \end{aligned}$$

Theorem of the square

$$t_{x+y}^*L \otimes L \simeq t_x^*L \otimes t_y^*L$$

$$\rightarrow (t_{x+y}^*L \otimes L^{-1}) \simeq (t_x^*L \otimes L^{-1}) \otimes (t_y^*L \otimes L^{-1})$$

Thus

$$\begin{aligned} \phi_L : X &\rightarrow \text{Pic}^0(X) \\ x &\mapsto T_x^*L \otimes L^{-1} \text{ is homomorphism.} \end{aligned}$$

Fact TFAE ($L = L(D) : D : \text{eff div}$)

- (1) $K(L) : \text{finite}$
- (2) $|2D|$ bpf. $\nexists$ finite morph $X \rightarrow \mathbb{P}^N$
- (3) L is ample on X

idea of pf of (1) $\Rightarrow$ (3)

$|2D|$ contains $t_x^*D + t_{-x}^*D$ (by theorem of the square)

For any $p \in X$, $\begin{cases} (1) p \notin \text{Supp}(D) \Rightarrow p \notin 2D \Rightarrow p \notin |2D| \\ (2) p \in \text{Supp}(D) \Rightarrow \exists x \in X \text{ s.t. } p \pm x \in \text{Supp}(D) \end{cases}$

$$\int_{D-x} \int_p \int_{D+x} \bigcap_{\mathbb{P}^1} \mathbb{P}^1$$

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Thus $Bs|2D| = \emptyset$ $\#$

Fact $|3D|$: very ample

We can find an eff div D s.t $H = \{x \in X \mid t_x^* D = D\}$ is finite

$$\begin{cases} U \subset X \text{ affine open} \\ D_1, \dots, D_t : \text{components of } X - U \\ D = \sum D_i \end{cases}$$

H : complete & U : affine, then H is finite.

Therefore

Every abelian variety is projective

It's hard to find equation defining abel. var

→ using theta function (next time...)